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SUPERSEDING
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MILITARY SPECIFICATION

FUZE, ROCKET, BASE DETONATING XM434 ASSEMBLING AND PACKING

This specification is approved for use by the U.S. Army Missile Command, Department of the Army, and is available for use by all Departments and Agencies of the Department of Defense.

1. SCOPE

1.1 This specification contains requirements not normally required by the drawings and provides the quality assurance provisions for the fabrication of parts, assembly and packing of one type of fuze designated as Fuze, Rocket: Base Detonating, XM434.

2. APPLICABLE DOCUMENTS

* 2.1 Issues of documents. The following documents of the issue in effect on date of invitation for bids or request for proposal form a part of this specification to the extent specified herein.

SPECIFICATIONS

MILITARY

MIL-A-2550	Ammunition and Special Weapons; General Specification for,
MIL-P-14137	Primer, Stab M106 Loading, Assembling and Packing

STANDARDS

MILITARY

MIL-STD-105	Sampling Procedures and Tables for Inspection by Attributes
MIL-STD-109	Quality Assurance Terms and Definitions

FSC 1340

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MIL-STD-331	Environmental and Performance Tests for Fuze and Fuze Components
MIL-STD-650	Explosive, Sampling, Inspection Testing
MIL-STD-1168	Lot Numbering of Ammunition
MIL-STD-1169	Packaging; Packing and Marking for Shipment of Inert Ammunition Components
MIL-STD-1235	Single or Multilevel Continuous Sampling Procedures and Tables for Inspection by Attributes

DRAWINGS

ARMY

8798862 -Carton, Packing, Ammunition for Fuze, Grenade, Rifle, T1014E1 or T1022E2, or Fuze, Rocket, 66MM, M412 without Booster Pellet, Cup Retaining Washer and Cup Disc

8798863 -Box, Packing, Ammunition for Fuze, Grenade, Rifle, T1014E1 or T1022E2, or Fuze, Rocket, 66MM, M412 without Booster Pellet, Cup Retaining Washer and Cup Disc.

9252815 -Fuze, Rocket: BD, XM434

PUBLICATIONS

ARMY

ET-9252815 -Equipment Tabulation

(Copies of specifications, standards, drawings and publications required by contractors in connection with specific procurement functions should be obtained from the procuring activity or as directed by the contracting officer.)

3. REQUIREMENTS

3.1 Materials. Materials shall be in accordance with applicable drawings and specifications.

3.1.1 Parts. All parts used in the assembly of the fuze shall meet all requirements of applicable drawings and specifications.

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3.2 Fuze, less primer cord. The fuze, less primer cord, shall comply with all requirements specified on Drawing (dwg.) 9252815, and with all requirements specified in applicable specifications. All reference to fuze or complete fuze in this specification shall be considered as the complete fuze less primer cord.

3.3 Spring embrittlement. Springs used in the fuze assembly shall be free from hydrogen embrittlement as evidenced by not breaking when extended to the point of minimum permanent distortion.

3.4 Porosity of mounting base. Porosity in the sidewall of the firing pin hole shall not exceed the following requirements:

Area A: Area A is described as the first 0.375 inch of the firing pin hole as measured from the plug end of the base. The total porosity of all voids in this area shall not exceed 0.007 square inch (0.095 diameter).

Area B: Area B is described as a length of 0.095 inch along the length of the firing pin hole adjacent to Area A. The total porosity of all voids in the area shall not exceed 0.0007 square inch (0.030 diameter).

Area C: Area C is described as the remainder of the length of the firing pin hole. The total porosity of all voids in this area shall not exceed 0.0013 square inch (0.040 diameter).

3.5 Electrical resistance.

3.5.1 Setback assembly. The electrical resistance between the telltale inner contact and the setback support shall not exceed ten ohms when all leaves are in the unarmed position. The electrical resistance between these same two points shall be not less than one megohm measured when all leaves are in the armed position.

3.5.2 Cover assembly. The electrical resistance between the telltale rivet and the cover shall be not less than one megohm.

3.5.3 Complete fuze. The maximum electrical resistance, indicative of an unarmed fuze, shall not exceed ten ohms when measured between the telltale contact in the cover assembly and the cover.

3.6 Operation of setback assembly.

3.6.1 Non-functioning. The setback assembly, in the base plate and rotor assembly, shall not function when subjected to a minimum arming acceleration of 800 gravities (g's) for a minimum of 5 seconds. Functioning

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is defined as movement of the leaves sufficient to permit leaf number 1 to clear the rotor pin.

3.6.2 Functioning time. The functioning time of the setback assembly, in the base plate and rotor assembly, shall not exceed 5.5 milliseconds while the base plate and rotor assembly is being subjected to maximum arming acceleration of 1,100 g's. Functioning time is defined as the interval from the release of leaf number 3 until leaf number 1 has moved to clear the rotor pin. Upon completion of the functioning time test, the rotor shall be in the fully armed position.

3.7 Fuze functioning. The complete fully armed fuze shall not function the explosive elements when subjected to a minimum acceleration of 5 g's for a minimum of 5 seconds, directed towards the cup end of the fuze and along the longitudinal axis of the fuze. The complete fully armed fuze shall function the explosive elements when the acceleration is increased to a maximum of 35 g's.

3.8 Rotor arming time.

3.8.1 At plus 70 degrees Fahrenheit (F.). The rotor assembly, in the base plate and rotor assembly, shall rotate from the full safe position to the fully armed position in not less than 70 milliseconds or more than 90 milliseconds.

3.8.2 At minus 40 degrees F. The rotor assembly, in the base plate and rotor assembly, shall rotate from the full safe position to the fully armed position in not less than 70 milliseconds or more than 100 milliseconds.

3.9 Jolt and jumble. The fuze shall be subjected to, and must withstand, the jolt test specified in Standard MIL-STD-331, Test 101, with the exception that the fuze shall not be disassembled and inspected after the jolt test. After completion of the jolt test the same fuze shall be subjected to and withstand the jumble test specified in Standard MIL-STD-331, Test 102.

* 3.10 Rough handling. The fuze shall function satisfactorily after being subjected to the transportation vibration specified in Standard MIL-STD-331, Test 104 and the five foot drop test specified in Standard MIL-STD-331, Test 111, except that the fuze shall be dropped bare, Impact shall be on the base end with the base as level as possible. The fuze shall satisfy the criteria for Passing Test as described in MIL-STD-331.

3.11 Workmanship. All assemblies shall be fabricated in a thorough, workmanlike manner and shall be free of cracks, burrs or sharp edges. All parts shall be free of chips, dirt, grease, rust and other foreign material

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that may impair proper functioning. The cleaning method shall not be injurious to any of the parts nor shall the parts be contaminated by the cleaning agents. Exterior surface coatings shall be continuous except for a few light scratches not exposing base material. All required markings shall be neat, clean and sharply defined.

3.12 First article inspection. This specification makes provisions for first article inspection. Requirements for the submission of first article samples by the contractor shall be specified in the contract.

3.13 Moisture content of detonator assembly explosives.

3.13.1 Lead azide. The moisture content of the lead azide used in loading the detonator, at the time of loading, shall not exceed 0.50 percent.

3.13.2 Pentaerythrite tetranitrate (PETN). The moisture content of the PETN used in loading the detonator, at the time of loading, shall not exceed 0.50 percent.

3.14 Waterproofness of detonator. The detonator shall be waterproof as evidenced by normal functioning after being immersed in water maintained at a temperature of 20 degrees Centigrade (C.), plus or minus 1 degree C., for a period of 48 hours minimum, at a depth of 2 to 3 inches.

3.15 Functioning of detonator. The detonator shall function on the first attempt when initiated by an M106 Primer and shall cause a 1/4 inch minimum diameter perforation in a 0.1345 inch minimum thickness lead disc when tested as specified herein.

4. QUALITY ASSURANCE PROVISIONS

4.1 Responsibility for inspection. Unless otherwise specified in the contract, the contractor is responsible for the performance of all inspection requirements as specified herein. Except as otherwise specified in the contract, the contractor may use his own or any other facilities suitable for the performance of the inspection requirement specified herein, unless disapproved by the Government. The Government reserves the right to perform any of the inspections set forth in the specification where such inspections are deemed necessary to assure supplies and services conform to the prescribed requirements. Reference shall be made to MIL-STD-109 in order to define terms used herein. The provisions of MIL-A-2550 shall apply.

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4.1.1 Submission of product. At the time each completed lot of items deliverable under the contract is submitted to the Government for acceptance, the contractor shall supply the following information accompanied by a certificate which attests that the information provided is correct and applicable to the product being submitted:

a. A statement that the lot complies with all the quality assurance provisions specified in this specification.

b. Drawing and specification number and date, together with identification and date of changes thereto.

c. A statement that all material purchased by the contractor meets requirements, when such material is controlled by Government or commercial specifications referenced in any of the contractual documents, and that certificates of conformance are on file and available for review, except that statements of findings on the moisture content of the lead azide and PETN shall be furnished with each lot submitted for acceptance.

d. Number of items in the lot.

e. Date submitted.

The certificate shall be signed by a responsible agent of the certifying organization. The initial certificate submitted shall be substantiated by evidence of the agent's authority to bind his principal. Substantiation of the agent's authority will not be required with subsequent certificates unless, during the course of the contract, this authority is vested in another agent of the certifying organization.

4,2 First article inspection.

* 4.2.1 Submission. The contractor shall submit a first article sample as designated by the contracting officer for evaluation in accordance with the provisions of 4.2.2. The first article sample shall consist of 60 complete fuzes, 125 detonator assemblies (dwg. 9240943), 15 base plate and rotor assemblies less primer (dwg. 9252817), 15 sets of subassemblies (each set of subassemblies shall consist of every subassembly covered by the drawing package) and 15 complete sets of component parts (each set shall consist of adequate components to fabricate a complete fuze) which have been produced by the contractor or furnished by a supplier and which have been manufactured using the same production processes, procedures and equipment which will be used in fulfilling the contract. All parts and materials, including packaging and packing, shall be obtained from the same sources of supply as will be used in regular production. Prior to submission, the contractor shall inspect the sample to the degree necessary to assure that it conforms to the requirements of the contract

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and submit a record of this inspection with the sample, including statements of findings of tests and examinations and certificates of conformance for materials, (see 4.1.1.c). A sample containing known defects will not be submitted unless specifically authorized by the contracting officer. A first article sample, or portion thereof, as directed by the contracting officer, shall also be submitted whenever there is a lapse in production for a period in excess of 90 days or whenever a change occurs in manufacturing process, material used, drawing, specification or source of supply such as to significantly affect product uniformity as determined by the Government.

4.2.2 Inspections to be performed. Assemblies and components will be subjected by the Government to any or all of the examinations or tests specified in 4.3.2 and 4.3.3 of this specification and any or all requirements of the applicable drawings. Destructive tests specified shall be performed only after all other inspections and tests for other characteristics have been completed.

4.2.3 Rejection. The first article sample shall be rejected if one or more fuzes, subassemblies or components fail to conform to any or all of the applicable requirements. The Government reserves the right to terminate its inspection upon any failure of a fuze, subassembly or component in the sample to comply with any of the specified requirements. In the event of rejection, the Government reserves the right to require the contractor to take corrective action and submit a new first article sample or portion thereof. Until a first article sample is accepted, the contractor is, in no way, authorized by the Government to initiate regular production (including procurement of material by the contractor) unless otherwise directed by the Contracting Officer.

4,3 Inspection provisions.

4.3,1 Lot formation. The term "inspection lot" as used in this specification is defined as an essentially homogeneous collection of units of product from which a representative sample is drawn and inspected to determine conformance with applicable requirements. The sample selected shall represent only that quantity of units from which the sample was drawn and shall not be construed to represent any prior or subsequent quantities presented for inspection. Homogeneity shall be considered to exist provided the inspection lot has been produced by one manufacturer, in one unchanged process, in accordance with the same drawing, same drawing revision, same specification and same specification revision and complies with the provisions for submission of product as specified in MIL-STD-105. Changes to the process, specification, or drawing not affecting safety, performance, interchangeability} or storage, as determined by the Government, shall not be deemed to alter the homogeneity of am inspection lot. All material submitted for inspection in accordance with this specification shall comply with the homogeneity

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criteria specified herein regardless of the type of sampling procedure which is being applied to determine conformance with requirements. Lot numbering, as required, shall be in accordance with MIL-STD-1168. Each inspection lot shall contain:

a. Parts from not more than one interfix lot number from one manufacturer.

b. Lead azide of the same type from not more than two lots produced by the same manufacturer in accordance with the same specification and same specification revision and which have subsequently been subjected to uniformly controlled drying and operating conditions at the loading plant.

c. PETN from not more than one lot.

d. Primers from not more than one interfix lot number from one manufacturer.

4.3.2 Examination. The examinations listed in the following sub-paragraphs shall be performed on inspection lots (see 4.3.1) of the components or assemblies designated in the applicable paragraph headings. Inspection for critical defects (and major defects, when so specified) shall be 100 percent. Sampling plans for major and minor defects shall be in accordance with MIL-STD-105 except that continuous sampling plans in accordance with MIL-STD-1235 may be used if approved by the Government. Instead of applying the AQL's designated herein to the groups of major and minor defects listed, the contractor may elect to apply AQL's to individual defects using a AQL of 0.40 percent for each major defect and an AQL of 0.65 percent for each minor defect. When this method is elected, it shall be applied to all of the major and minor defects listed in the applicable subparagraph of this section, except where 100 percent inspection is specified, and be documented as part of the contractor's inspection system, Equipment necessary for the performance of the inspections listed shall be in accordance with 4.3.4.

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4.3.2.1 Spring, leaf No. 2 (see dwg. 8796414 covering a detail of dwg. 10048781).

Categories	Defects	Method of Inspection	Code No. (see 6.2)
Critical:	None defined.		
Major:	AQL 0,40 percent		
101.	Inside diameter, minimum (min)..	Gage	01001
102.	Protective finish incomplete or inadequate	Visual	01002
Minor:	AQL 0.65 percent		
201.	Evidence of poor workmanship (see 3.11)	Visual	01003

4.3.2.2 Spring, leaf No. 3 (see dwg. 8796415 covering a detail of dwg. 10048781).

Categories	Defects	Method of Inspection	Code No.
Critical:	None defined.		
Major:	AQL 0.65 percent		
101.	Torque at angular displacement. . .	Gage	02001
102.	Inside diameter, min.	Gage	02002
103.	Protective finish incomplete or inadequate	Visual	02003

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Categories	Defects	Method of Inspection	Code No.
Minor:	AQL 0.65 percent		
201.	Evidence of poor workmanship (see 3.11)	Visual	02004

4.3.2.3 Spring, rotor (see dwg. 10048581 covering a detail of dwg. 9252817).

Categories	Defects	Method of Inspection	Code No.
Critical:	None defined.		
Major:	AQL 0.65 percent		
101.	Torque improper at either or both angular displacements	Gage	03001
102.	Inside diameter, min	Gage	03002
103.	Solid height, maximum (max.)	Gage	03003
104.	Protective finish incomplete or inadequate	Visual	03004
Minor:	AQL 0.65 Percent		
201.	Evidence of poor workmanship (see 3.11)	Visual	03005

4.3.2.4 Spring, firing pin (see dwg. 8796380 covering a detail of dwg. 9252815).

Categories	Defects	Method of Inspection	Code No.
Critical:	None defined.		
Major:	AQL 0.65 Percent		
101.	Solid height, max	Gage	04001
102.	Load at assembled height	Gage	04002
103.	Protective finish incomplete or inadequate.	Visual	04003
Minor:	AQL 1.00 percent		
201.	Ends not ground.	Visual	04004
202.	Evidence of poor workmanship (see 3.11)	Visual	04005

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4.3.2.5 Pin, firing (see dwg. 8796383 covering a detail of dwg. 9252815)

Categories	Defects	Method of Inspection	Code No.
Critical:	None defined.		
Major:	AQL 0.65 percent		
101.	Profile of firing pin point. , ...	Gage	05001
102.	Outside diameter, min	Gage	05002
103.	Inside diameter, max	Gage	05003
104.	Protective finish incomplete or inadequate	Visual	05004
Minor:	AQL 1.00 percent		
201.	Concentricity of inside diameter with outside diameter	Gage	05005
202.	Evidence of poor workmanship. . (see 3.11)	Visual	05006

4.3.2,6 Leaf No. 1 (see dwg. 8796408 covering a detail of dwg. 10048781).

Categories	Defects	Method of Inspection	Code No.
Critical:	None defined.		
Major:	AQL 0.40 percent		
101.	Plating on inside diameter of hole incomplete or damaged	Visual	06001
102.	Finish improper	Visual	06002
Minor:	AQL 0.65 percent		
201.	Evidence of poor workmanship (see 3.11)	Visual	06003

4.3.2.7 Leaf No. 2 (see dwg. 8796410 covering a detail of dwg. 10048781.

Categories	Defects	Method of Inspection	Code No.
Critical:	None defined.		
Major:	AQL 0.40 percent		
101.	Plating in inside surface of hole incomplete or damaged	Visual	07001
102.	Finish improper	Visual	07002

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Categories	Defects	Method of Inspection	Code No.
Minor:	AQL 0.65 percent		
201.	Evidence of poor workmanship (see 3.11)	Visual	07003

4.3.2.8 Leaf No. 3 (see dwg. 10048783 covering a detail of dwg. 10048781).

Categories	Defects	Method of Inspection	Code No.
Critical:	None defined.		
Major:	AQL 0.40 percent		
101.	Plating on inside surface of stud hole incomplete or damaged	Visual	08001
102.	Finish improper	Visual	08002
Minor:	AQL 0.65 percent		
201.	Evidence of poor workmanship (see 3.11)	Visual	08003

4.3.2.9 Stud, leaf No. 1 (see dwg. 8796409 covering a detail of dwg. 10048781.

Categories	Defects	Method of Inspection	Code No.
Critical :	None defined.		
Major:	AQL 0.40 percent		
101.	Plating on intermediate diameter incomplete or damaged	Visual	09001
102.	Finish improper	Visual	09002
Minor:	AQL 0.65 percent		
201.	Evidence of poor workmanship (see 3.11)	Visual	09003

4.3.2.10 Stud leaf, No. 2 and No. 3 (see dwg. 8796411 and 8796412 covering a detail of dwg.

Categories	Defects	Method of Inspection	Code No.
Critical:	None defined.		
Major:	AQL 0.40 percent		
101.	Plating incomplete	Visual	10001

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Categories	Defects	Method of Inspection	Code No.
	102. Finish improper	Visual	10002
Minor:	AQL 0.65 percent		
	201. Evidence of poor workmanship (see 3.11)	Visual	10003

4.3.2.11 Base, mounting (see dwg. 10048784 covering a detail of dwg. 10048785).

Categories	Defects	Method of Inspection	Code No.
Critical:	None defined.		
Major:	AQL 0.65 percent		
	101. Height to cover seating surface, min	Gage	11001
	102. Height to top of cylindrical portion of base, max	Gage	11002
	103. Finish of firing pin cavity improper	Visual	11003
	104. Finish of locking shaft cavity improper	Visual	11004
Minor:	AQL 0.65 percent		
	201. Evidence of poor workmanship (see 3.11)	Visual	11005

4.3.2.12 Setback assembly (see dwg. 10048781 covering a detail of dwg. 9252817).

Categories	Defects	Method of Inspection	Code No.
Critical:			
	1. Stud leaf No. 1 and/or stud leaf No. 2 not peened secure	Visual	12001
*	2. Electrical resistance between setback support and telltale inner contact (see 3.5.1)	Test (see 4.4.6)	12002
	3. Distance between the front surface of the set back support and the rear surface of leaf number 1 above, max	Gage	12003

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Categories	Defects	Method of Inspection	Code No.
Major:	AQL 0.65 percent		
101.	Telltale inner contact loose or deformed.	Visual/Manual	12004
102.	Telltale insulator projects beyond aft surface of telltale inner contact . . .	Visual	12005
103.	Plating damaged (other than peened students).	Visual	12006
104.	Assembly incorrect or incomplete	Visual	12007
Minor:	AQL 0.65 percent		
201.	Evidence of poor workmanship (see 3.11)..	Visual	12008

4.3.2.13 Rotor assembly (see 9252818 covering a detail of dwg. 9252817).

Categories	Defects	Method of Inspection	Code No.
Critical:			
1.	Push test of rotor pin....	Test	13001
Major:	AQL 1.00 percent		
101.	Plating damaged other than stake marks . . .	Visual	13002
102.	Detonator washer improperly staked	Visual	13003
104.	Rotor gear improperly staked	Visual	13005
106.	Flash hole between primer cavity and detonator cavity obstructed	Visual	13007
107.	Components missing or loose	Visual/Manual	13008
Minor:	AQL 0.65 percent		
201.	Evidence of poor workmanship (see 3.11)	Visual	13009

4.3.2.14 Base plate and rotor assembly (see dwg. 9252817 covering a detail of dwg. 9252815).

Categories	Defects	Method of Inspection	Code No.
Critical:	None defined.		
Major:	AQL 1.00 percent		
101.	Primer inverted or loose	Visual/Manual	14001
102.	Staking inadequate or improperly	Visual	14002
103.	Components missing or damaged	Visual	14003

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Categories	Defects	Method of Inspection	Code No.
104.	Rotor slot not clinched over rotor spring	Visual	14004
105.	Retaining spring improperly positioned damaged, and/or not under tension (see dwg. 10048785)	Manual . .. Visual/Manual	14005
106.	Protruding burr in primer staking area.	Visual	14016
Minor:	AQL 0.65 percent		
201.	Evidence of poor workmanship (see 3.11)	Visual	14006

4.3.2.15 Cover assembly (see dwg. 9252816 covering a detail of dwg. 9252815).

Categories	Defects	Method of Inspection	Code No.
Critical:			
1.	Electrical resistance between telltale rivet and cover (see 3.5.2)	Test (see 4.4.4)	15001
Major:	AQL 1.00 percent		
101.	Cup not concentric with outside diameter of cover	Gage	15002
102.	Cover plage not concentric with outside diameter of cover.	Gage	15003
103.	Flash hole obstructed by excessive brazing material	Visual	15004
104.	Plating incomplete or inadequate	Visual	15005
105.	Telltale rivet loose	Manual	15006
Minor:	AQL 1.50 percent		
201.	Depth of cup, min	Gage	15007
202.	Window(s) damaged obscuring vision . .	Visual	15008
203.	Evidence of poor workmanship (see 3.11)	Visual	15009

4.3.2,16 Fuze, Rocket, BD XM434, prior to assembly of primer cord (see dwg. 9252815).

Categories	Defects	Method of Inspection	Code No.
Critical:			
1.	Fuze in the armed or partially armed position (see 3.5.3)	Test (see 4.4.5)	16001

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2.	Electrical Test mark missing (telltale contact-cover)	Visual (see 4.4.5)	16002
3.	Rotor in armed or partially armed position	Visual (see 4.4.5)	16017
Major:	AQL 0.40 percent		
101.	Base plug loose or above flush	Manual/Manual	16003
102.	Assembly dented or damaged	Visual	16004
Minor:	AQL 1.00 percent		
201.	Marking incorrect, incomplete, illegible or missing	Visual	16005
202.	Evidence of poor workmanship (see 3.11)	Visual	16006

4.3.2.17 Cup, (see dwg. 9252743 covering a detail of dwg. 9240943).

Categories	Defects	Method of Inspection	Code No.
Critical:			
1.	Split or cut	Visual	17001
Major:	AQL 0.65 percent		
101.	Thickness through bottom, min	Gage	17002
102.	Outside diameter, max	Gage	17003
103.	Dent or distortion	Visual	17004
Minor:	AQL 1.00 percent		
201.	Inside diameter, min	Gage	17005
202.	Evidence of poor workmanship (see 3.11)	Visual	17006

4.3.2.18 Plug (see dwg. 9252737 covering a detail of dwg. 9240943).

Categories	Defects	Method of Inspection	Code No.
Critical:	None defined.		
Major:	AQL 0.40 percent		
101.	Outside diameter, min	Gage	18001
102.	Diameter of sleeve surface, min	Gage	18002
Minor:			
201.	Evidence of poor workmanship (see 3.11)	Visual	18003

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4.3.2.19 Sleeve (See dwg. 9252744 covering. a detail of dwg. 9240943).

Categories	Defects	Method of Inspection	Code No.
Critical:	None defined.		
Major:	None defined.		
Minor:	AQL 1.50 percent		
201.	Outside diameter, max.	Gage	
	Length, max	Gage	19002
203.	Evidence of poor workmanship (See 3.11)	Visual	19003

4.3.2.20 Plug, Assembly (se dwg. 9252738 covering a detail of dwg. 9240943).

Categories	Defects	Method of Inspection	Code No.
Critical:	None defined.		
Major:	None defined.		
Minor:	AQL 1.50 percent		
201.	Length, max	Gage	20001
202.	Sleeve loose	Manual	20002
203.	Evidence of poor workmanship (see 3.11)	Visual	20003

4.3.2.21 Assembly, Plug Loading, prior to applying priming charge (see dwg. 9252739 covering a detail of dwg. 9240943).

Categories	Defects	Method of Inspection	Code No.
Critical:	None defined		
Major:	AQL 0.40 percent		
101.	Flash charge missing	Visual	21001
Minor:	AQL 0.65 percent		
201.	Evidence of poor workmanship (see 3.11)	Visual	21002

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4.3.2.22 Assembly, Plug Loading (see dwg. 9252739 covering a detail of dwg. 9240943).

Categories	Defects	Method of Inspection	Code No.
Critical:			
	1. Explosive on exterior	Visual	22001
Major:	AQL 0.40 percent		
	101. Priming charge missing	Visual	22002
Minor:	AQL 0.65 percent		
	201. Evidence of poor workmanship (see 3.11)	Visual	20003

4.3.2.23 Detonator Assembly (see dwg. 9240943 covering a detail of dwg. 9252818).

Categories	Defects	Method of Inspection	Code No.
Critical :			
	1. Cup split or cracked	Visual	23001
Major:	AQL 1.00 percent		
	101. Plug cracked	Visual	23002
	102. Total length, max.	Gage	23003
	103. Diameter of cup at plug end, max	Gage	23004
	104. Crimp not full 360 degrees	Visual	23005
	105. Varnish missing or inadequate	Visual	23006
Minor:	AQL 1.00 percent		
	201. Diameter of cup at charge end, max . .	Gage	23007
	202. Evidence of poor workmanship (see 3. 11)	Visual	23008

4.3.2.24 Carton, packing (see dwg. 8798862).

Categories	Defects	Method of Inspection	Code No.
Critical:	None defined.		
Major:	AQL 0.40 percent		
	101. Sealing strip loose, torn, incomplete, wrinkled, or improperly adhered	Visual	24001

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Minor:	AQL 1.50 percent		
201.	Contents move when shaken	Manual	24003
202.	Label missing.	Yisual	24004
203.	Marking incorrect, incomplete, illegible, or missing	Visual	24005

4.3.2.25 Box, packing (see dwg. 8798863).

Categories	Defects	Method of Inspection	Code No.
Critical:	None defined.		
Major:	AQL 0.65 percent		
101.	Top improperly assembled	Visual/Manual	25001
102.	Board broken or split	Visual/Manual	25002
103.	Strapping missing, broken or loose . .	Visual/Manual	25003
104.	Contents exposed	Visual	25004
Minor:	AQL 1.00 percent		
201.	Contents loose	Manual	25005
202.	Marking misleading or unidentifiable. .	Visual	25006

4.3.3 Testing

4.3.3.1 First article sample (see 4.2.1). The samples submitted as the first article quantity shall be tested as specified herein after being examined 100 percent for the applicable classifications of defects in paragraph 4.3.2.

4.3.3.1.1 Solid firm lubricant (see dwg. 8796419, 8796420 and 10048602) - Major defect - Star Wheel - CodeNo.26001; Pallet - Code No. 26002; and Pinion gear - Code No. 26003. A sample of 15 of each component shall selected for the first article and tested as specified in 4.4.1. Rejection shall be as specified in 4.2.3.

4.3.2.1.2 Spring embrittlement (see 3.3). Critical defect - Spring leaf No. 2 - Code No. 01005; Spring leaf No. 3 - Code No. 02005; Major defect - Spring rotor - Code No. 03006 and and Spring firing pin - Code No. 04006. A sample of 15 springs shall be selected of each spring type for the first article and tested as specified in 4.4.2. Rejection shall be as specified in 4.2.3.

4.3.3.1.3 Porosity and surface finish of mounting base (see 3.4 and dwg. 10048784) - Porosity - Major defect - Code No, 11006; Surface finish - Major defect - Code No. 11007. A sample of 15 mounting bases shall be selected for the first article and tested as specified in 4,4.3. Rejection shall be as specified in 4.2.3.

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4.3.3.1.3.1 Surface finish of graze lever shaft of mounting base (see dwg. 10048784) - Major defect - Code No. 11008. A sample of 15 mounting bases shall be selected for the first article and tested as specified in 4.4.3.1. Rejection shall be as specified in 4.2.3.

4.3.3.1,4 Operation of setback assembly (see 3.6); Functioning during non-functioning test - Critical defect - Code No. 14007; Failure to function within maximum time on functioning time test - Major defect - Code No. 14008. A sample of 15 base plate and rotor assemblies shall be selected for the first article and tested as specified in 4.4,8 using test equipment specified in 4.3.4. Rejection shall be as specified in 4.2.3.

4.3.3.1.5 Functioning of fuze elements (see 3.7) - Major defect - Code No. 14009; Examination for positioning of locking shaft (see dwg. 9252817) - Major defect - Code No. 14010. A sample of 15 base plate and rotor assemblies (less primer) shall be selected for the first article and assembled with a firing pin, firing pin spring, and base plug to each assembly. The assembly shall then be tested as specified in 4.4.7 using test equipment specified in 4.3.4. Visual examination shall be performed under three power magnification to assure that the locking shaft is bearing properly on the graze lever. Assemblies which contain locking shafts which do not bear on the graze lever as illustrated on dwg. 9252817, shall be classed defective. Rejection shall be as specified in 4.2.3.

4.3.3.1,6 Rotor arming time at plus 70 degrees Fahrenheit (F.) - (see 3.8.1) - Arming time, minimum - Critical defect - Code No. 14011; and arming time, maximum - Major defect - Code No. 14012. A sample of 15 base plate and rotor assemblies shall be selected for the first article and tested as specified in 4.4.9 using test equipment specified in 4.3.4. The temperature conditioning limitations shall be 65°F. to 75°F. Rejection shall be as specified in 4.2.3.

* 4.3.3.1.7 Environmental and destructive testing. A sample of 60 fuzes shall be selected for the first article and subjected to the following tests in the sequence listed.

* 4.3.3. 1.7.1 Jolt and jumble (see 3.9): Failure to withstand test - critical defect - Code No. 16012; Missing plate screws - Major defect - Code No. 16007. Ten of the 60 fuzes selected in 4.3.3.1.7. shall be assembled into closures or simulated closures. The assembly shall then be subjected to the jolt test specified in Standard MIL-STD-331, Test 101 and then without dis-assembly to the jumble test specified in Standard MIL-STD-331, Test 102. The first article shall be rejected if, after disassembly, one or more fuzes have functioned or if one or more fuzes are armed or unsafe to handle, or if two or more plate screws have become completely disengaged.

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* 4.3.3.1.7.2 Rough Handling (see 3.10). The remaining 50 fuzes shall be subjected to the transportation-vibration test and then without disassembly to the five-foot drop test specified in 3.10.

* 4.3.3.1.7.2.1 Arming Tests. These tests shall be performed on thirty (30) of the fifty (50) fuzes only. The cover assembly graze elements and base plug components shall be removed but the graze elements (firing pin and firing pin spring) identified for later re-insertion into the same fuze and the fuzes shall be subjected to the following tests in the sequence specified using test equipment specified in 4.3.4.

TEST	TEST PROCEDURE
Non-functioning of setback assembly	4.4.8
Functioning time of setback assembly	4.4.8
Rotor arming time at plus 70 degrees F	4.4.9
Rotor arming time at minus 40 degrees F	4.4.9

The temperature tolerance adhered to during performance of the rotor arming time test at plus 70 degrees F and minus 40 degrees F shall be plus and minus 5 degrees F.

The first article sample shall be rejected if one or more defective are found according to the defects listed in Table I below. The results may be examined and tested for outliers, but only one outlier is allowed and only for the minus 40 degrees F temperature. Refer to 6.6 and 6.7 for calculations of sample standard deviation (sigma) and outliers respectively. After completion of testing (see 4.4.9), the cover assemblies, graze elements and base plugs shall be reassembled to the fuzes and the fuzes tested as follows in 4.3.3.1.7.2.2.

TABLE I
ARMING TEST DEFECTIVES

	Defect	Category	Code No.
1.	Functioning during non-functioning test of setback assembly (see 3.6.1)	Critical	16008
2.	Functioning time of setback assembly exceeds maximum (see 3.6.2)	Major	16009
3.	Rotor arming time @ 70 degrees F less than minimum (see 3.8.1)	Critical	16010
4.	Rotor arming time @ 70 degrees F more than maximum (see 3.8.1)	Major	16011

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	Defect	Category	Code No.
5.	Rotor arming time @ minus 40 degrees F less than minimum (see 3.8.2)	Critical	16015
6.	Rotor arming time @ minus 40 degrees F more than maximum (see 3.8.2)	Major	16016
7.	Rotor fails to arm @ either temperature	Major	16020
8.	Average rotor arming time plus 1.90 sigma exceeds maximum specified time @ either temperature	Major	16018
9.	Average rotor arming time minus 1.90 sigma is less than the minimum time specified @ either temperature	Major	16019
10.	Assembly fails rough handling treatment (see 3.10)	Major	16022

* 4.3.3.1 .7.2.2 Explosive graze functioning (See 3.7). The fifty (50) fuzes previously tested in 4.3.3.1 .7.2 and 4.3.3.1.7.2.1 shall be subjected to the test specified in 4.4.10 using test equipment specified in 4.3.4. The first article sample shall be rejected if one or more fuzes are classed defective according to the defects defined in Table II. Fuzes which do not function shall not be included in the calculations for average graze functioning level, and outliers are not permitted. Refer to 6.6 for calculation of sample standard deviation (sigma).

TABLE II

EXPLOSIVE GRAZE FUNCTIONING TEST DEFECTIVES

	Defect	Category	Code No.
1.	Fuze fails to arm @ maximum acceleration level (see 3.6.2)	Major	16020
2.	Fuze functions during non-functioning graze acceleration test (see 3.7)	Major	16013
3.	Fuze fails to function explosively or explosive elements fail to perforate the booster cup	Major	16021

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4.	Average graze functioning level plus or minus 1.60 sigma fails to satisfy the specified requirements (see 3.7)	Major	16014

4.3.3. 1.8 Determination of moisture content.

4.3.3.1 .8.1 Lead azide (see 3.13.1) - Major Defect - Code No. 27001. The moisture content of the lead azide shall be determined for the first article sample at the time of loading. A sample of 2 grams shall be selected and tested as specified in 4.4.11. If the moisture content is in excess of the specified percentage, any detonators already loaded shall be rejected. A first article sample of detonator assemblies shall not be submitted until proper corrective action has been accomplished as verified by successfully repeating the moisture content test as specified.

4.3.3.1 .8.2 PETN (see 3.13.2) - Major defect - Code No. 27002. The moisture content of the PETN shall be determined for the first article sample at the time of loading. A sample of 5 grams shall be selected and tested as specified in 4.4.12. If the moisture content is in excess of the specified percentage, any detonators already loaded shall be rejected. A first article sample of detonator assemblies shall not be submitted until proper corrective action has been accomplished as verified by successfully repeating the moisture content test as specified.

4.3.3.1.9 Waterproofness and functioning of detonator assembly (see 3.14 and 3.15). Detonator fails to function - Major defect - Code No. 23009: Inadequate perforation of lead disc - Major defect - Code No. 23010. - A sample of 125 detonator assemblies (dwg. 9240943) shall be selected for the first article and tested as specified in 4.4.13 using test equipment as specified in 4.3.4. Rejection shall be as specified in 4.2.3.

4.3.3.2 Production.

4.3.3.2.1 Solid film lubricant (see dwg. 8796419, 8796420 and 10048602) - Major defect - Star wheel - Code No. 26001; Pallet - Code No. 26002; and Pinion gear - Code No. 26003. A sample of 5 of each component shall be selected from each lot and tested as specified in 4.4.4. The component lot shall be rejected if one or more components are classed defective.

4.3.3.2.2 Spring embrittlement (see 3.3) - Critical defect Spring leaf No. 2 - Code No. 01005; Spring leaf No. 3 - Code No. 02005; Major defect - Spring rotor - Code No. 03006 and Spring firing pin - Code No. 04006. A sample of ten springs shall be selected from each spring type included in each heat treatment batch and tested as specified in 4.4.2. If one or more springs are classed defective, the heat treatment batch shall be rejected.

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4.3.3.2.3 Porosity and surface finish of mounting base (see 3.4 and dwg. 10048784) - Porosity - Major defect - Code No. 11006 ; Surface Finish - Major defect - Code No. 11007. A sample of 10 mounting bases shall be selected from each lot and tested as specified in 4.4.3. If two or more mounting bases fail to conform to the requirements specified, the lot shall be rejected.

4.3.3.2.3.1 Surface finish of graze lever shaft and mounting base (see dwg. 10048784) - Major defect - Code No. 1008. A sample of 10 mounting bases shall be selected from each lot and tested as specified in 4.4.3.1. If one or more mounting bases is classed defective, the lot shall be rejected.

4.3.3.2.4 Operation of setback assembly (see 3.6); Functioning during non-functioning test - critical defect - Code No. 14007; Failure to function within maximum time on functioning time test - Major defect - Code No. 14008. This test shall be performed 100 percent on base plate and rotor assemblies. The test shall be performed as specified in 4.4.8 using test equipment specified in 4.3.4. Defective assemblies shall be rejected and removed from the lot.

* 4.3.3.2.5 Functioning of fuze elements (see 3.7) non-functioning, functioning - Major defect - Code No. 14009; Examination for positioning of locking shaft (see dwg. 9252817) - Major defect - Code No. 14010. This test and examination shall be performed 100 percent on base plate and rotor assemblies, less primer, (see dwg. 92528170 by assembling a firing pin, firing pin spring, and base plug to each assembly. The assembly shall then be tested as specified in 4.4.7 using test equipment specified in 4.3.4. Assemblies classed defective may be disassembled and the firing pin or the firing pin spring or the locking shaft or by combination thereof may be replaced and retested as often as necessary to obtain acceptable assemblies. Assemblies which contain locking shafts which do not bear on the graze level as illustrated on the applicable drawing shall be classed defective and removed from the lot, subject to repositioning of the locking shaft and re-examination prior to re-entry into the lot. The set of graze elements shall be identified so that subsequent assembly of the graze elements into the proper base plate and rotor assemblies is assured.

* 4.3.3.2.6 Rotor arming time at plus 70 degrees Fahrenheit (°F) - (see 3.8.1) - Arming time, minimum - Critical defect - Code No. 14011; and Arming time, maximum - Major defect - Code No. 14012. his test shall be performed 100 percent on base plate and rotor assemblies conditioned within the temperature limitations of 65°F. and 85°F., and the assemblies shall comply with the requirements as specified in paragraph 3.8.1. The test shall be performed as specified in 4.4.9 using test equipment specified in 4.3.4. Assemblies failing to arm in the time specified shall be classed defective and removed from the lot. On completion of arming time test visually confirm rotor restored to unarmed position - Critical defect - Code No. 14017.

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4.3.3.2,7 Environmental and destructive testing. A sample of 60 fuzes shall be selected from each lot and subjected to the following testing in the sequence listed.

* 4.3.3.2.7.1 Jolt and Jumble (see 3.9) - Critical defects - functioning and arming - Code No. 16012; and missing plate screws (see 3.9) - Major defect - Code No. 16007. Ten of the 60 fuzes selected in 4.3.3.2.7 shall be assembled into closures or simulated closures. The assembly shall then be subjected to the jolt test specified in MIL-STD-331, Test 101 and then without disassembly to the jumble test specified in MIL-STD-331, Test 102. The lot shall be rejected if, after disassembly, one or more fuzes have functioned or if one or more fuzes are armed or unsafe to handle, or if two or more plate screws have become completely disengaged.

* 4.3.3.2.7.2 Rough handling (see 3.10). The remaining 50 fuzes shall be subjected to the transportation vibration test and then without disassembly to the five-foot drop test specified in 3.10.

* 4. 3.3.2.7.2.1 Arming tests: These tests shall be performed on thirty (30) of the fifty (50) fuzes only. The cover assembly, graze elements and base plug components shall be removed, but the graze elements (firing pin and firing pin spring) identified for later re-insertion into the same fuze; and the fuzes shall be subjected to the following tests in the sequence specified using test equipment specified in 4.3.4.

TEST	TEST PROCEDURE
Non-functioning of setback assembly	4.4.8
Functioning time of setback assembly	4.4.8
Rotor arming time at plus 70 degrees F	4.4.9
Rotor arming time at minus 40 degrees F	4.4.9

The temperature tolerance adhered to during performance of the rotor arming time test at plus 70 degrees F and minus 40 degrees F shall be plus and minus 5 degrees F.

The lot shall be rejected if one or more defective are found according to the defects listed in Table I (see 4.3.3.1.7.2.1). The results may be examined and tested for outliers but only one outlier is allowed and only for the minus 40 degree F temperature. Refer to 6.6 and 6.7 for calculations of sample standard deviation (sigma) and outliers respectively. After completion of testing (see 4.4.9), the cover assemblies, graze elements and base plugs shall be reassembled to the fuze and the fuzes tested as follows in 4,3.3.2.7.2.2.

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* 4.3.3.2.7.2.2 Explosive graze functioning (see 3.7). The fifty (50) fuzes previously tested in 4.3.3.2.7.2 and 4.3.3.2.7.2.1 shall be subjected to the test specified in 4.4.10 using test equipment specified in 4.3.4. The lot shall be rejected if two (2) or more fuzes are classed defective according to the defects defined in Table II (see 4.3.3.1.7.2.2). Fuzes which do not function shall not be included in the calculations for average graze functioning level and outliers are not permitted. Refer to 6.6 for calculation of sample standard deviation (sigma).

4.3.3.2.8 Check test for possible deterioration (see MIL-P-14137 and 6.5). If the total elapsed time between the original acceptance of any initiator lot and the assembly of that lot into the fuze exceeds two years, or if the initiators have been subjected to adverse conditions for any period of time since previous tests, the initiator lot shall be subjected to and must satisfactorily pass the check test for possible deterioration required by the initiator specification immediately before the initiator lot is assembled into the fuze. These tests shall be performed by the contractor on initiators selected by the Government inspector at the facility assembling the initiators into the fuze. The test shall be conducted at Government expense without cost to the contractor who loaded the initiators or the contractor assembling the initiators into the fuze and shall not constitute a basis for rejection against either contractor except where deterioration has occurred as a direct result of carelessness in handling, storage, etc., permitted while the initiator lot was under the jurisdiction of either contractor.

4.3.3.2.9 Heat seal test of sealed bag (see dwg. 8798862) - Major defect- Code no. 17002. Sampling, acceptance criteria and test procedures for this test shall be as required in MIL-P-116 (Preservation, Methods of). This test is applicable only when level B packing is required.

4.3.3.2.10 Determination of moisture content.

4.3.3.2.10.1 Lead azid (see 3.13.1) - Major defect - Code No. 27001. The moisture content of the lead azide shall be determined for each lot at the time of loading. A sample of 2 grams shall be selected and tested as specified in 4.4.11. If the moisture content is in excess of the specified percentage and loading has not begun, the lot of lead azide shall be rejected until proper corrective action has been accomplished as verified by successfully repeating the moisture content test as specified. If loading has begun, action on the unloaded lead azide shall be as specified above and all detonators loaded previously from the same lot of lead azide shall be rejected. The contractor shall maintain adequate records to provide identification of the lead azide used in the loaded detonators.

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4.3.3.2.10.2 PETN (see 3.13.2) - Major defect - Code No. 27002. The moisture content of the PETN shall determined for each lot at the time of loading. A sample of 5 grams shall be selected and tested as specified in 4.4.12. If the moisture content is in excess of the percentage specified and loading has not begun, the lot of PETN shall be rejected until proper corrective action has been accomplished as verified by successfully repeating the moisture content test as specified. If loading has begun, action on the unloaded PETN shall be as specified above and all detonators loaded previously from the same lot of PETN shall be rejected. The contractor shall maintain adequate records to provide identification of the PETN used in the loaded detonators.

4.3.3.2.11 Waterproofness and functioning of detonator assembly (see 3.14 and 3.15). Detonator fails to function - Major defect - Code No. 23009; Inadequate perforation of lead disc - Major defect - Code No. 23010. A sample of 3.15 detonator assemblies shall be selected at random from each lot and tested as specified in 4.4.13 using test equipment as specified in 4.3.4. If threee or more assemblies fail to comply with the requirements of 3.14 and 3.15, the lot shall be rejected. If one or two defects are found, a second sample of 315 detonator assemblies shall be selected at random from the same lot and tested as above. If, in the combined first and second sample, a total of four or more defects are found, the lot shall be rejected.

* 4.3.3.2.12 Inspection for proper assembly of graze elements - Locking shaft bearing improperly on graze lever - Major defect - Code No. 14017. This inspection shall be performed 100% on each complete base plate and rotor assembly (dwg. 9252817) after assembly of the firing pin (dwg. 8796383), firing pin spring (dwg. 8796380) and base plug (dwg. 9796384) and immediately prior to installation of cover assembly (dwg. 9252816). Assemblies shall be examined in accordance with 4.4.14. Defective shall be removed from the lot and subjected to repositioning of the locking shaft. If the graze elements are removed or disturbed on any assembly, the inspection shall be repeated prior to completing the fuze assembly (dwg. 9252815).

4.3.4 Inspection equipment. Equipment Tabulation ET-9252815 identifies the inspection equipment required to perform the examinations and tests prescribed in this section. The contractor shall design inspection equipment in accordance with the instructions in 6.3.

4.3.4.1 Government rights to documentation. Inspection equipment drawings and lists provided and revised in accordance with the requirements of the ET may be used by DoD activities for design, procurement, manufacture, testing, evaluation, production and receiving inspection, overhaul, shipping, storage, identification of stock, ordering and storage of replacement parts, inspection of items at overhaul, general maintenance of equipment, construction, survey and whenever inspection equipment drawings are needed.

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4.4 Test methods and procedures.

4.4.1 Solid film lubricant. The component shall be cross sectioned and the exposed surface polished. The film thickness shall be measured. The component shall be classed defective if bare spots exist on the surfaces or if the thickness of film exceeds the specified requirement.

4.4.2 Spring embrittlement. The spring shall be extended in a manner representative of actual functioning (i.e., bending, twisting, tension) until the spring no longer returns to its original condition. The spring shall be classed defective if it breaks prior to this point.

4.4.3 Porosity of mounting base. The mounting base shall be sectioned through the center of the firing pin hole, parallel to the locking shaft hole. The firing pin hole shall be examined for porosity and surface finish. The mounting base shall be classed defective if porosity is in excess of the specified requirements or if the surface finish exceeds the specified requirements.

4.4.3.1 Surface finish of graze lever shaft of mounting base. The mounting base shall be sectioned through the center of the graze lever shaft and each of the sections shall be examined for surface finish using the equipment specified in 4.3.4. The mounting base shall be classed defective if it fails to comply with the specified surface finish requirement.

4.4.4 Insulation resistance. The electrical resistance of the cover assembly shall be measured between the points specified in 3.5.2 using the test equipment specified in 4.3.4. The cover assembly shall be classed defective if the resistance measurement is less than the specified requirement.

* 4.4.5 Fuze safety. The electrical resistance shall be measured as specified in 3.5.3. A red dot shall be readily visible on the cover plate to indicate that the fuze has passed the telltale contact-cover electrical test. The test equipment (see 4.3.4) shall automatically stamp the proper marking on each fuze, only when each electrical resistance measurement is within the required limits. The fuze shall be closed defective and removed from the lot if after having been subjected to the test the acceptance mark is not present on the fuze. Prior to performance of the fuze safety test, the unarmed position of the rotor shaft shall be visually confirmed (see dwg. 9252815).

WARNING: This test may be dangerous and should be performed behind a suitable barricade.

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4.4.6 Electrical resistance of setback assembly. The electrical resistance shall be measured between the points specified in 3.5.1 using test equipment specified in 4.3.4. The test shall be performed with the leaves in each of the specified positions. The assembly shall be classed defective if one or both resistance measurements fail to meet the specified requirements.

4.4.7 Graze functioning. The assembly shall be examined to assure proper positioning of the locking shaft and graze level. Prior to testing the assembly shall provide means to insure against damage to the firing pin and the rotor when the firing pin is release during the functioning phase of this test. The armed assembly shall be mounted on a horizontal centrifuge with the longitudinal axis of the assembly lying along a path of radial acceleration, and the base end pointed towards the center of rotation. The center of gravity is defined as a point 0.50 inches from the base end of the assembly. The minimum non-functioning acceleration shall be applied and maintained for the minimum time period specified. Observation shall be made for non-functioning or functioning. If the assembly has not functioned, the acceleration shall be increased until functioning occurs or until the maximum functioning acceleration is attained. The assembly shall be classed defective if it functions (firing pin is released) during the non-functioning phase or if it fails to function during the functioning phase.

4.4.8 Operation of setback assembly.

* 4.4.8,1 Non-functioning. The assembly shall be mounted on a horizontal centrifuge with the longitudinal axis of the assembly lying along a path of radial acceleration, and the base end pointed away from the center of rotation. The minimum non-arming acceleration shall be applied for the minimum time specified and observation made for functioning or non-functioning. The assembly shall be classed defective if the setback assembly functions during the non-functioning phase. The center of gravity of the assembly is defined as a point 0.35 inches from the base end of the assembly.

4.4.8.2 Functioning time. The assembly shall be mounted as in 4.4.8.1. The test equipment shall lock leaf number 3 in the fully safe position and the required acceleration applied to the assembly and stabilized. The number 3 leaf shall be released and the time interval measured from release of leaf number 3 until leaf number 1 moves to a position that would release the rotor. The assembly shall be classed defective if the measured time interval exceeds the specified requirement or if the rotor fails to be in the fully armed position after the arming acceleration has been removed.

WARNING: This test may be dangerous and should be performed behind a suitable barricade.

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4.4.9 Rotor arming time. The assemblies shall be conditioned within the specified temperature for a minimum of 3 hours. The complete test shall be performed at the required temperature. The arming time (with all leaves in the fully armed position) shall be measured from when the rotor is released from the fully safe position (rotor pin against end of slot) until the rotor reaches the fully armed position, as indicated when the primer cavity in the rotor is in line with the firing pin cavity in the base of the fuze. The time shall be recorded and the assembly returned to the fully unarmed condition.

WARNING: This test may be dangerous and should be performed behind a suitable barricade.

* 4.4.10 Explosive graze non-functioning and functioning. The assembly shall be placed in the test fixture and subjected to the required arming acceleration. The assembly shall then be observed for proper arming (failures shall be removed without further testing) and re-oriented to permit performance of graze testing. The armed assembly shall then be subjected to the minimum non-functioning graze acceleration specified for the minimum time period specified. If functioning does not occur, the acceleration shall be increased until functioning occurs and the functioning g-level shall be recorded to determine compliance with the applicable requirements.

WARNING: This test is dangerous and should be performed behind a suitable barricade.

4.4.11 Moisture content of lead azide. The moisture content of the lead azide shall be determined in accordance with Method 101.5 of MIL-STD-650 except that a two gram sample shall be tested and the dish shall be covered with a sheet of polytetrafluoroethylene.

4.4.12 Moisture content of PETN. The moisture content of the PETN shall be determined according to the method specified in 4.4.11 for lead azide except that a five gram sample shall be used and the drying time shall be one hour.

4.4.13 Waterproofness and functioning of detonator assembly. prior to the functioning test the detonator assembles shall be conditioned for forty-eight (48) hours immersed in water at a depth of two (2) to three (3) inches and maintained at a temperature of twenty (20) degrees Centigrade (C), plus or minus one (1) rotor assembly (dwg. 9252818), or its equivalent, and fired by the

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flash of an M106 Primer, see MIL-P-14137, which has also been assembled in the rotor assembly in accordance with dwg. 9252817. Explosive output of each detonator assembly shall be measured by the diameter of the perforation in a 0.1345 inch minimum thickness lead disc. The M106 Primer shall be initiated as specified in MIL-P-14137. Detonator assemblies which fail to fire (duds) or which fail to achieve the required perforation diameter in the lead disc shall be classed defective.

* 4.4.14 Proper assembly of graze elements. Visually examine the exposed end of the locking shaft, under three power magnification, for proper positioning as shown in dwg. 9252817. Any assembly in which the locking shaft is not bearing on the end of the graze level, or in which the locking shaft is improperly oriented, shall be classed defective.

5. PREPARATION FOR DELIVERY

5.1 Preservation, packaging, packing and marking.

5.1.1 Level B. The fuze, less primer cord, shall be packaged, packed and marked in accordance with dwg. 8798862 and 8798863 including the barrier bag listed on 8798862.

5.1.2 Level C.

5.1.2.1 Fuzes. The fuzes, less primer cord, shall be packaged, packed and marked as specified in 5.1.1 except that barrier bags will not be required.

5.1.2.2 Unmatches, unassembled parts. If the fuzes are shipped with the graze elements not assembled, the base plugs shall be packaged, packed, and marked in accordance with MIL-STD-1169. For explosive components, suitable commercial packaging to meet Hazardous Materials Regulations of the Department of Transportation shall be used.

6. NOTES

6.1 Ordering data. Procurement documents shall specify the following:

a. Title, number and date of this specification.

b. Ammunition data cards shall be prepared for each lot in accordance with MIL-STD-1167.

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6.2 Inspection code numbers. The five digit code numbers assigned to the inspections herein are to facilitate future data collected and analysis by the government. All submitted designs shall contain a reference to the applicable 5 digit code number contained in Section 4 of this specification and the appropriate component or assembly drawing number and revision letter to which the specific design applies. These code numbers are used to correlate the characteristics cited on Equipment Lists with the inspections listed in this specification. They should also be cited as references on drawings of equipment designs submitted by the contractor (see 6.3.5).

6.3 Inspection equipment. The contractor shall design inspection equipment as required by the referenced inspection Equipment Lists (EL) in accordance with the instructions of paragraphs 6.3.1 through 6.3.5.

6.3.1 Inspection equipment lists. Inspection equipment lists indicate the availability of Inspection equipment designs by showing in the "number" column of the list of inspection equipment (Form SMUPA 1010) the numbers of drawings or Federal Stock Numbers of existing equipment designs, or codes as indicated in paragraph 6.3.2. Design action required of the contractor is described in paragraphs 6.3.3 and 6.3.4. The contractor will be required to prepare detailed drawings in accordance with 6.3.4 for all the equipment coded as "contractor design" in the number column. These contractor designs must be approved by the Government prior to fabrication or procuring of the equipment. Designs shall be submitted for approval as specified in 6.3.5.

6.3.2 Inspection equipment list codes. The inspection equipment as defined in 6.3.3 and 6.3.4 will be designated in the EL by the following codes:

CD - Contractor design on controlled contractor format and/or commercial equipment.

MU - Army design, mandatory for use.

6.3.3 Army designs. Army designs are reflected on detailed drawings which completely depict all the information necessary for the fabrication of the item of inspection equipment. The contractor need provide no design when an Army design is listed for an item of inspection equipment. Army designs fall into two basic classifications; mandatory (designated "MU") and non-mandatory (indicated by drawing or Federal Stock Number). When an inspection equipment list references mandatory Army designs, the

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contractor shall comply with, and use these designs accordingly. The contractor may, however, in connection with non-mandatory designs, and with the approval of the Government, design alternate inspection equipment or use comparable commercial equipment to facilitate his operations. Such contractor prepared designs or commercial equipment selections must be approved by the government prior to fabrication or procuring of the equipment. Designs shall be submitted for approval as specified in 6.3.5.

6.3.4 Contractor designs. Contractor designs are designs of inspection equipment for which the Government has assigned design responsibility to the contractor. Contractor designs shall be supported by detailed drawings which depict all information necessary to completely fabricate, calibrate and operate an item of inspection equipment. This requires that the necessary views, dimensions, materials, finish, notes, operating and calibration instructions be properly depicted in accordance with approved practices to the extent that further calculation or clarification will not be required. Unless otherwise specified, contractor designs may be developed on the format the contractor normally employs in his equipment design procedure provided such format reflects the detail and information specified above, subject to the following controls: All submitted contractor designs shall conform to MIL-D-1000, Category E, Form 2. Legibility and reproducibility shall permit conventional marking of clearly understandable, high contrast reproductions. Contractors shall submit one copy of final designs as a rolled set. Designs shall be submitted for approval as specified in 6.3.5.

* 6.3.5 Submission of contractor design. All submitted designs shall contain a reference to the applicable five digit Code number contained in Section 4 of this specification. Unless otherwise specified on the EL, all designs of equipment for inspection of defects classified as critical and major shall be submitted for approval to the Government in accordance with the appropriate contract. All other designs of inspection equipment shall be approved by the inspection element of the agency administering the contract; submission shall be as directed by the contracting officer. Partial submission of inspection equipment designs is permissible and encouraged. However, the Arsenal completion date for design review will be based on the date of the final submission of designs.

* 6.4 Submission of first article samples. Instructions as to location for the evaluation of the first article shall be obtained from the contracting officer. All inquiries should be forwarded to the activity designated in the contract.

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6.5 Testing of government owned initiators. The contracting officer will arrange to have the fuze contractor reimbursed for expenses incurred in the performance of the check for deterioration of detonators and primers when the contractor is directed to perform the tests.

6.6 Calculations of sample standard deviations. The sample standard deviation shall be calculated using the formula as follows:

$$S = \sqrt{\frac{\Sigma(X - \bar{X})2}{n - 1}}$$

Where: S = sample standard deviation
X = numerical value for individual observations
$\bar{X}$ = average value for the sample
n = sample size

6.7 Calculation of outliers. In order to test for the presence of a mathematical outlier in a given set of test data the following procedure will apply. Identify the highest value of arming time in the set of test data as the suspected outlier and calculate a sample average and sample standard deviation excluding the high value from the computation. Then calculate a value for "K" using the following equation:

$$K = \frac{\text{High Value-Average}}{\text{Standard Deviation}}$$

If the calculated value of "K" exceeds 2.9, then the high value is an outlier.

6.8 Changes to previous issue. The margins of this specification are marked to indicate where changes (additions, modification, corrections, deletions) from the previous issue were made. This was done as a convenience only and the Government assumes no liability whatsoever for any inaccuracies in these notations. Bidders and contractors are cautioned to evaluate the requirements of this document based on the entire content irrespective of the marginal notations and relationship to the last previous issue.

Custodian:
Army - MI

Preparing Activity:
Army - MI

Review Activity:
Army - PA

Project No. 1340-A499

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STANDARDIZATION DOCUMENT IMPROVEMENT PROPOSAL

OMB Approval
No. 22-R255

INSTRUCTIONS: The purpose of this form is to solicit beneficial comments which will help achieve procurement of suitable products at reasonable cost and minimum delay, or will otherwise enhance use of the document. DoD contractors, government activities, or manufacturers/vendors who are prospective suppliers of the product are invited to submit comments to the government. Fold on lines on reverse side, staple in corner, and send to preparing activity. Comments submitted on this form do not constitute or imply authorization to waive any portion of the referenced document(s) or to amend contractual requirements. Attach any pertinent data which may be of use in improving this document. If there are additional papers, attach to form and place both in an envelope addressed to preparing activity.

DOCUMENT IDENTIFIER AND TITLE

NAME OF ORGANIZATION AND ADDRESS

CONTRACT NUMBER

MATERIAL PROCURED UNDER A

☐ DIRECT GOVERNMENT CONTRACT ☐ SUBCONTRACT

1. HAS ANY PART OF THE DOCUMENT CREATED PROBLEMS OR REQUIRED INTERPRETATION IN PROCUREMENT USE?

A GIVE PARAGRAPH NUMBER AND WORDING

B RECOMMENDATIONS FOR CORRECTING THE DEFICIENCIES

2. COMMENTS ON ANY DOCUMENT REQUIREMENT CONSIDERED TOO RIGID

3. IS THE DOCUMENT RESTRICTIVE?

☐ YES ☐ NO *(If "Yes", in what way?)*

4. REMARKS

SUBMITTED BY *(Printed or typed name and address - Optional)*

TELEPHONE NO

DATE

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